

Sasha Sutton

Master's student – Data, Knowledge & Hybrid Artificial Intelligence (DKAI), Université Paris-Saclay | NLP & Computer Vision

+33 6 74 57 76 85 sashasuttons3@icloud.com portfolio linkedin.com/in/sashasutton4 github.com/sashsutton

Education

Université Paris-Saclay Master 1 in Computer Science – Data, Knowledge & Hybrid Artificial Intelligence (DKAI) - Selective AI track taught in English: machine learning, big data, symbolic & hybrid AI. Aix-Marseille University Bachelor of Science in Mathematics and Computer Science Relevant Coursework: Data Structures and Algorithms (Java), Probability & Statistics, Object-Oriented Programming (Java), Linear Algebra, Calculus - Completing a second bachelor's degree in Mathematics (L3, double-degree programme) alongside the M1 – expected June 2027. Middlesex University Bachelor of Arts in Music Production & Sound Engineering	Sept. 2026 – June 2027 Gif-sur-Yvette, France June 2026 Marseille, France October 2023 London, UK
---	--

Experience

Laboratory of Computer Science and Systems (LIS) Natural Language Processing (NLP) Intern - Developed Retrieval-Augmented Generation (RAG) systems by designing and implementing conversational AI pipelines for historical web archives from the BnF and INA. - Performed topic modelling and semantic theme extraction on large-scale heterogeneous web-archive (WARC) datasets to automate the discovery of historical trends and key themes. Micron Applied Dentistry SAS Statistical Data Analyst Intern - Built a Python data-processing pipeline to clean and aggregate 126 high-resolution 3D dental scans with noise filtering and statistical averaging. - Designed a reproducibility validation framework using pixel-wise standard deviation maps, spatial aggregation and a one-sided Student's t-test to verify sub-20 µm precision. - Developed an automatic anatomical landmark detection algorithm using cubic polynomial regression and second-derivative analysis to distinguish soft vs. hard tissue regions. - Performed power analysis and Monte Carlo simulations and produced publication-quality visualisations for PhD thesis validation reports.	Apr. 2026 – Jul. 2026 Marseille, France Sep. 2025 – Dec. 2025 Marseille, France
---	--

Research & Publications

“De la propagation avant à la rétropropagation : une analyse des mécanismes d'apprentissage dans les réseaux de neurones multicouches” – research article on the mathematical foundations of multilayer perceptrons (backpropagation, gradient descent, regularisation), Aix-Marseille Université, 2026. hal.science/hal-05628568

Projects

Neural Vision 3D – Interactive Machine Learning Visualizer React, Three.js, Python, FastAPI, NumPy, TypeScript - Developed an interactive 3D visualization of a Neural Network from scratch using React Three Fiber (Three.js), allowing users to inspect real-time neural activations in a 784-64-10 architecture. - Engineered the inference engine in Python/NumPy – manual forward pass with Sigmoid and Softmax activations, no high-level ML libraries – served via a FastAPI REST API for real-time recognition of hand-drawn digits. Machine Learning Library in C C, CMake - Implemented a lightweight neural-network library from scratch in C: matrix operations, dense layers, ReLU/Sigmoid activations and MSE loss, with manual memory management. - Structured the codebase as a reusable library (headers, sources, worked examples) built with CMake and GCC. Vector Search Engine (RAG Backend) Python, FastAPI, React, NumPy - Built a semantic search engine from scratch, implementing Cosine Similarity via matrix multiplication to rank documents by meaning rather than keywords, with Sentence-Transformers (BERT) 384-dimensional embeddings. - Deployed a full-stack microservices architecture: FastAPI backend on Render and React frontend on Vercel, connected via REST APIs. Air Harp – Computer Vision Musical Instrument Python, OpenCV, MediaPipe, Pygame - Built a real-time virtual instrument that tracks finger movements with MediaPipe hand-landmark detection and OpenCV, triggering low-latency multi-channel audio when fingers pluck virtual strings.
--

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, C, SQL (PostgreSQL, Supabase).
Frontend/3D: React.js, Next.js, Tailwind CSS, Shopify/Liquid, Bootstrap.
Backend/Cloud: Node.js, FastAPI, Express.js, Spring Boot, AWS (S3, EC2, Lambda), REST APIs.
AI & Data: NumPy, PyTorch, TensorFlow, Sentence-Transformers (BERT), OpenCV, MediaPipe, Pandas, Matplotlib.
Concepts: Retrieval-Augmented Generation (RAG), Topic Modelling, Neural Network Architectures, Vector Embeddings, Statistical Testing, Database Normalisation.

Social Engagements

Achievements: 1st Place Hackathon Winner (AMU 2026).
Memberships: Chess Club and Mathematics Club at Aix-Marseille University.
Interests & Sports: Mountain Cycling, Marathons, Kickboxing, Strategic Games.